

MODEL QUESTION PAPER

INDUSTRIAL ELECTRONICS &PLC

(Maximum Marks – 100)

Time- 3Hrs

PART - A

I Answer the questions in one or two sentences. Each question carries 2marks.

1. Draw the VI characteristics of SCR.
2. What is meant by natural commutation?
3. What is a DC chopper?
4. List four applications of induction heating.
5. What is a rotary encoder?

(5x2=10Marks)

PART - B

II Answer any five questions

(5x6=30 Marks)

1. Explain the SCR gate triggering method which gives a firing angle between 0-180 degrees.
2. Draw and explain the transfer characteristics of power MOSFET.
3. Draw light dimmer circuit using TRIAC and explain the operation of the circuit.
4. Explain the operation of a parallel inverter circuit.
5. Compare AC and DC motor drives.
6. Explain the principle of a servo controlled voltage stabilizer.
7. Draw a ladder diagram to turn on a lamp using two start switches, and to stop using any of the two stop switches.

PART - C

(Answer one full question from each unit)

UNIT - I

- IIIa). Draw structure and explain the VI characteristics of a DIAC. (7)
b). Draw the SCR pulse triggering circuit using UJT and explain the operation. (8)

OR

- IV a) Draw the structure and explain the operation of IGBT. (7)
b) Draw the structure and VI characteristics of TRIAC. (8)

UNIT - II

- V a) With relevant waveform and circuit explain the operation of a series inverter. (8)
b) Explain the operation of a single phase midpoint step up cycloconverter. (7)

OR

- VI a) Explain the operation of a Jone's chopper with circuit and waveforms (8)
b) Explain the operation of half wave converter with resistive load. Plot the waveforms across SCR and load. (7)

UNIT - III

- VII a) Draw and explain the speed control of dc series motor. (7)
b) List types of electric welding systems. Explain the spot welding system. (8)

OR

- VIII a) Draw block diagram of online UPS and explain operation. (7)
b) Explain the principle of dielectric heating. List any four applications of dielectric heating. (8)

UNIT – IV

- IX a) Write short notes on i) hall effect sensor ii) proximity switch (7)
b) Explain any four bit instruction. (8)

OR

- X a) Explain different functional blocks in a PLC (15)